

YAMAGoya

Open Source Threat Hunting Tool using YARA and SIGMA

Shusei Tomonaga
Tomoya Kamei



CODEBLUE2025

BECAUSE SECURITY MATTERS

Motivation

Endpoint protection products exist for malware detection, targeted attacks utilizing malware that operates only in memory remain challenging to detect using existing products and continue to pose a threat.

Endpoint protection to detect non-file-based activity depends on the black box of the security solution and cannot detect activities using legitimate tools used by the attackers.

The best way to threat hunting is to use YARA and SIGMA, and we need a security solution that uses those rules to endpoint detection.

We have created a new tool.



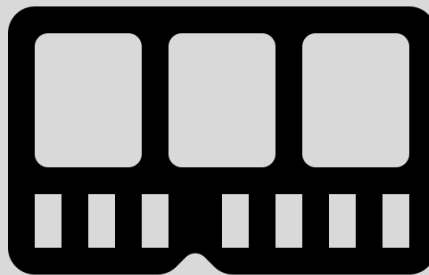


Create a threat hunting tool that can use YARA and SIGMA, knowledge of the security community.

Targets of This Tool



File

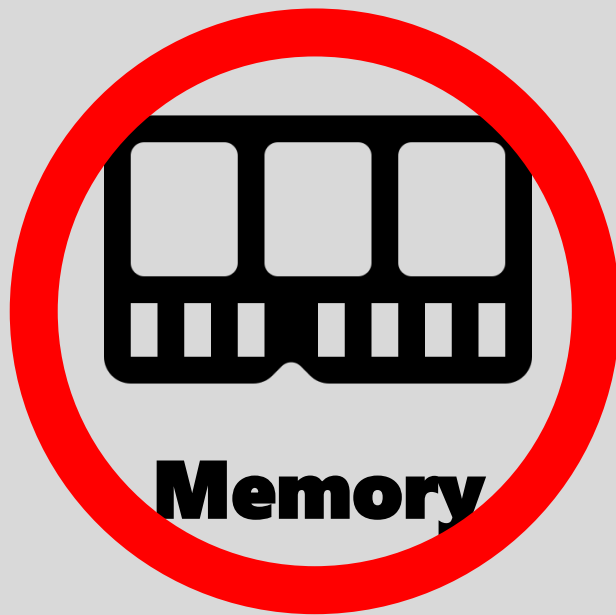


Memory



Behaviors

Targets of This Tool



This tool is not a file scanner.

What is YAMAGoya

 **YAMAGoya** is a C# application that leverages Event Tracing for Windows (ETW) to capture real-time system events and memory during threat hunting or incident response.



shu-tom Create .gitignore87b72e9 · last week🕒 3 Commits		
.github/workflows	Upload	last week
core	Upload	last week
images	Upload	last week
resource	Upload	last week
rules	Upload	last week
.gitignore	Create .gitignore	last week
App.xaml		
App.xaml.cs		
MainWindow.xaml	Upload	last week

About

Yet Another Memory Analyzer for malware detection and Guarding Operations with YARA and SIGMA

- Readme
- Activity
- Custom properties
- 0 stars
- 0 watching
- 0 forks

Releases

ublished
[release](#)

Packages

<https://github.com/JPCERTCC/YAMAGoya>

What is YAMAGoya



(Yet Another Memory Analyzer
for malware detection and
Guarding Operations with
YARA and SIGMA)

Features

- 1 Behavioral detection using **SIGMA** rules.
- 2 Detects malware using **YARA** rules and **memory information**.
- 3 Execute entirely in **userland**, avoiding kernel-mode dependencies.
- 4 Support for **GUI** and **CUI**.

Demo

- Malware detection using YAMAGoya -

C:\YAMAGoya>

Process Explorer - Sysinternals.com [TEST]i...

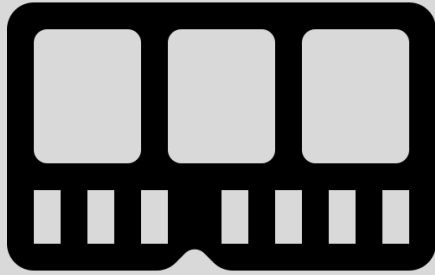
File Options View Process Find Users Help

<Filter by name>

Process	CPU	Private Bytes	Working Set	PID	Descripti
svchost.exe	1.696 K	10,020 K	2460 Host Pro		
svchost.exe	5,120 K	21,872 K	3040 Host Pro		
svchost.exe	2,716 K	13,112 K	3224 Host Pro		
svchost.exe	3,516 K	19,348 K	2320 Host Pro		
svchost.exe	4,740 K	14,692 K	3444 Host Pro		
svchost.exe	3,260 K	14,260 K	3904 Host Pro		
MoUsoCoreWorker.exe	16,540 K	39,420 K	1780 MoUSO C		
svchost.exe	7,744 K	28,696 K	5144 Host Pro		
svchost.exe	3,388 K	12,852 K	1404		
SearchIndexer.exe	12,520 K	27,064 K	1708 Microsof		
svchost.exe	2,112 K	12,056 K	5856 Host Pro		
svchost.exe	2,776 K	10,252 K	6304 Host Pro		
svchost.exe	2,988 K	14,368 K	6348 Host Pro		
SecurityHealthService.exe	< 0.01	4,360 K	18,464 K	6552	
svchost.exe	25,604 K	43,008 K	7316 Host Pro		
svchost.exe	1,768 K	8,064 K	3540 Host Pro		
svchost.exe	3,540 K	14,176 K	7848 Host Pro		
svchost.exe	1,792 K	10,288 K	8644 Host Pro		
MsMpEng.exe	< 0.01	270,048 K	150,000 K	8656	
MpDefenderCoreService.e...	10,092 K	24,240 K	8552		
NisSrv.exe	4,452 K	13,492 K	1824		
svchost.exe	8,496 K	30,136 K	6732 Host Pro		
svchost.exe	1,224 K	6,204 K	8892 Host Pro		
svchost.exe	1,700 K	8,396 K	312 Host Pro		
svchost.exe	6,712 K	12,148 K	12900 Host Pro		
svchost.exe	3,664 K	12,372 K	13056		
svchost.exe	10,148 K	23,000 K	9808 Host Pro		
svchost.exe	1,748 K	7,744 K	11872 Host Pro		
lsass.exe	7,996 K	25,360 K	844 Local Ser		
fontdrvhost.exe	1,352 K	4,340 K	976 Usermod		
explorer.exe	< 0.01	146,300 K	333,240 K	696 Windows	
SecurityHealthSystray.exe	1,956 K	12,484 K	6384 Windows		
vmtoolsd.exe	< 0.01	6,092 K	18,728 K	5952 VMware	
msedge.exe	47,380 K	133,208 K	1936 Microsof		
msedge.exe	2,108 K	11,440 K	6996 Microsof		
msedge.exe	13,280 K	43,344 K	4896 Microsof		
msedge.exe	11,608 K	30,840 K	6316 Microsof		
msedge.exe	8,184 K	21,992 K	5848 Microsof		
msedge.exe	78,388 K	129,248 K	7748 Microsof		
msedge.exe	14,824 K	32,344 K	7896 Microsof		
msedge.exe	12,484 K	25,488 K	9164 Microsof		
OneDrive.exe	< 0.01	70,108 K	137,848 K	7380 Microsof	
cmd.exe		3,168 K	5,292 K	9036 Windows	
conhost.exe					

CPU Usage: 0.50% Commit Charge: 20.62% Processes: 158 Physical Usage: 23.56%

Targets of This Tool



Memory

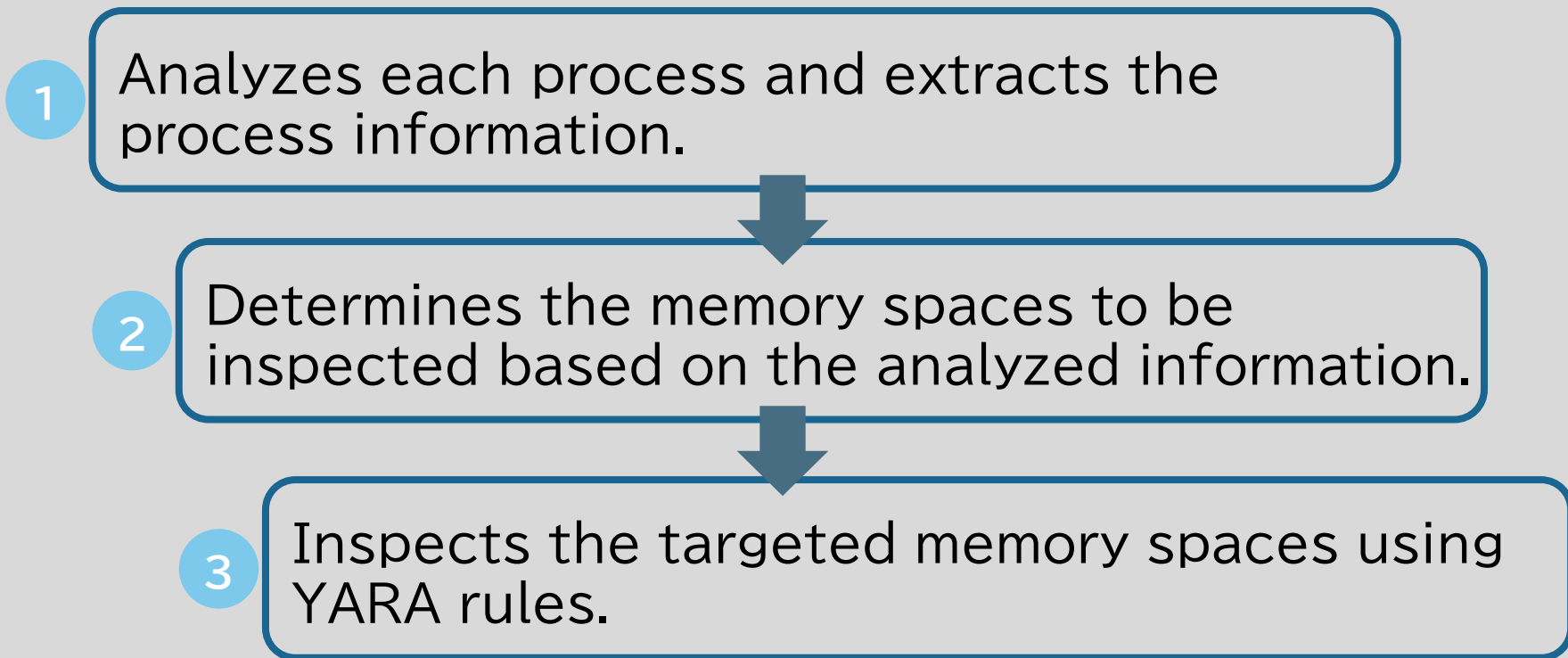


Behaviors

How it Works

 **Goya** operates as a Windows UserMode application, analyzing the memory spaces of processes running in user mode, scanning memory spaces with suspicious attributes using YARA, and identifying malware.

How it Works



Extract the Process information

1. Virtual Memory Map
2. Process PEB (Process Environment Block) structure
3. Process TEB (Thread Environment Block) structure
4. Process stack region
5. Process heap region
6. Process thread information

Tag the Memory Space to Scan

Win32 Process Memory Map

0x00000000	error trapping
0x00010000	
	program image
	heap
	stack
0x40000000	
	DLLs
	DLLs
	DLLs
0x7FFDE000	First TEB
0x7FFDF000	PEB
0x7FFE0000	Shared user page
0x7FFE1000	No access
0x7FFFFFFF	



Win32 Process Memory Map

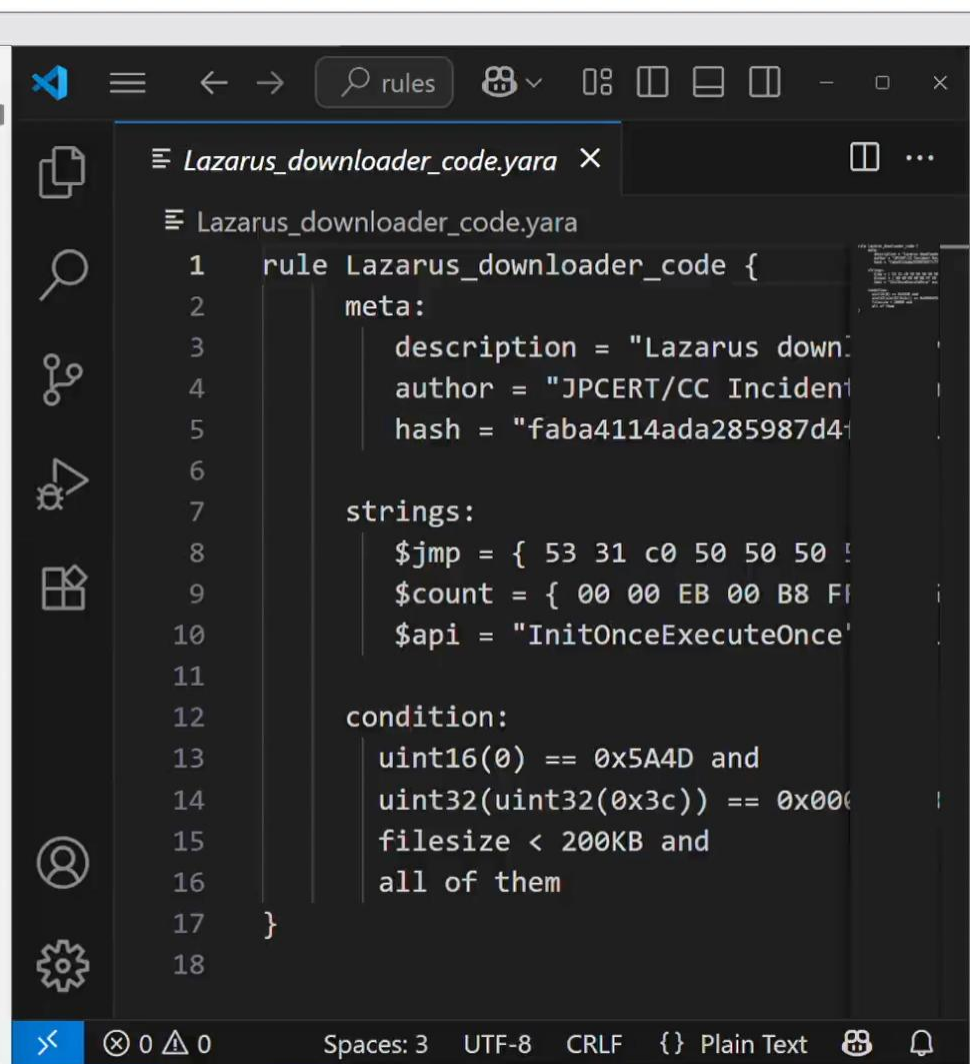
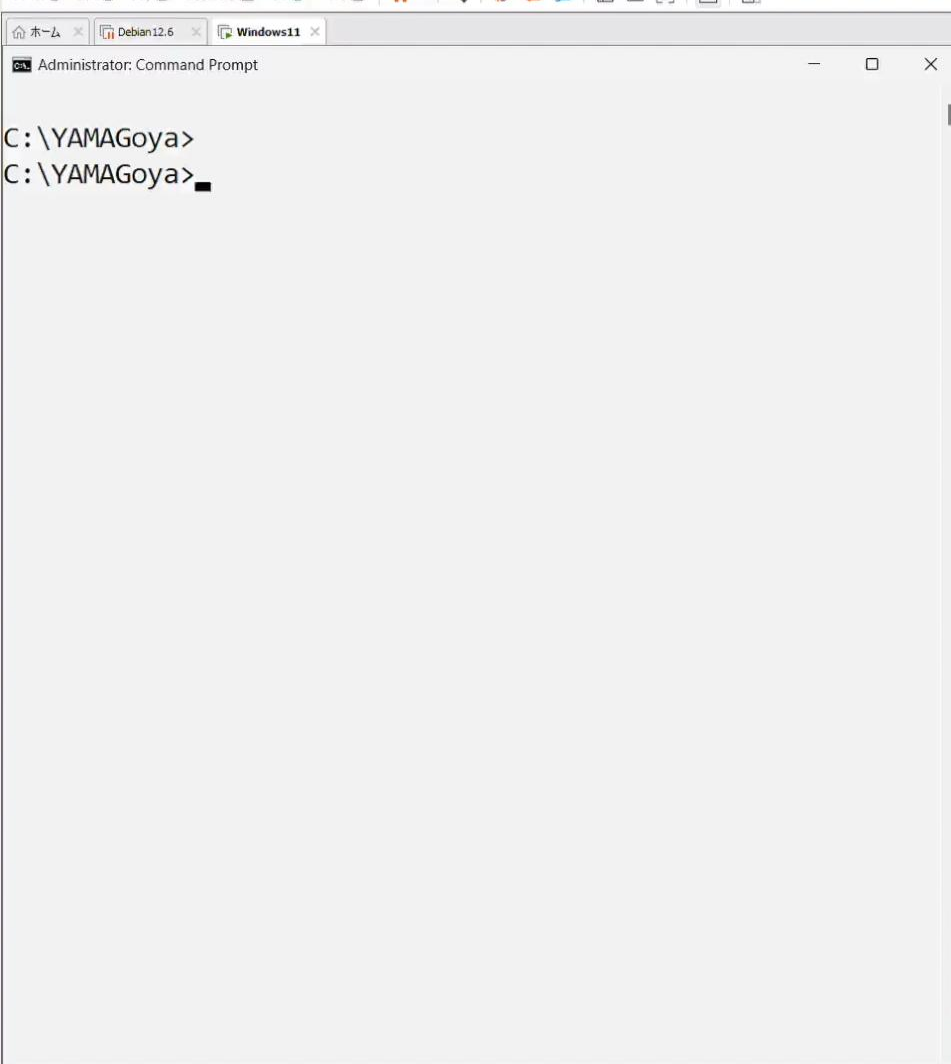
0x00000000	error trapping	Private
0x00010000		COMMIT
	program image	X
	heap	
	stack	
0x40000000		
	DLLs	
	DLLs	
	DLLs	
0x7FFDE000	First TEB	
0x7FFDF000	PEB	
0x7FFE0000	Shared user page	
0x7FFE1000	No access	
0x7FFFFFFF		



Scan with


Demo

- Memory scan using YARA rule -



Targets of This Tool



Memory



Behaviors

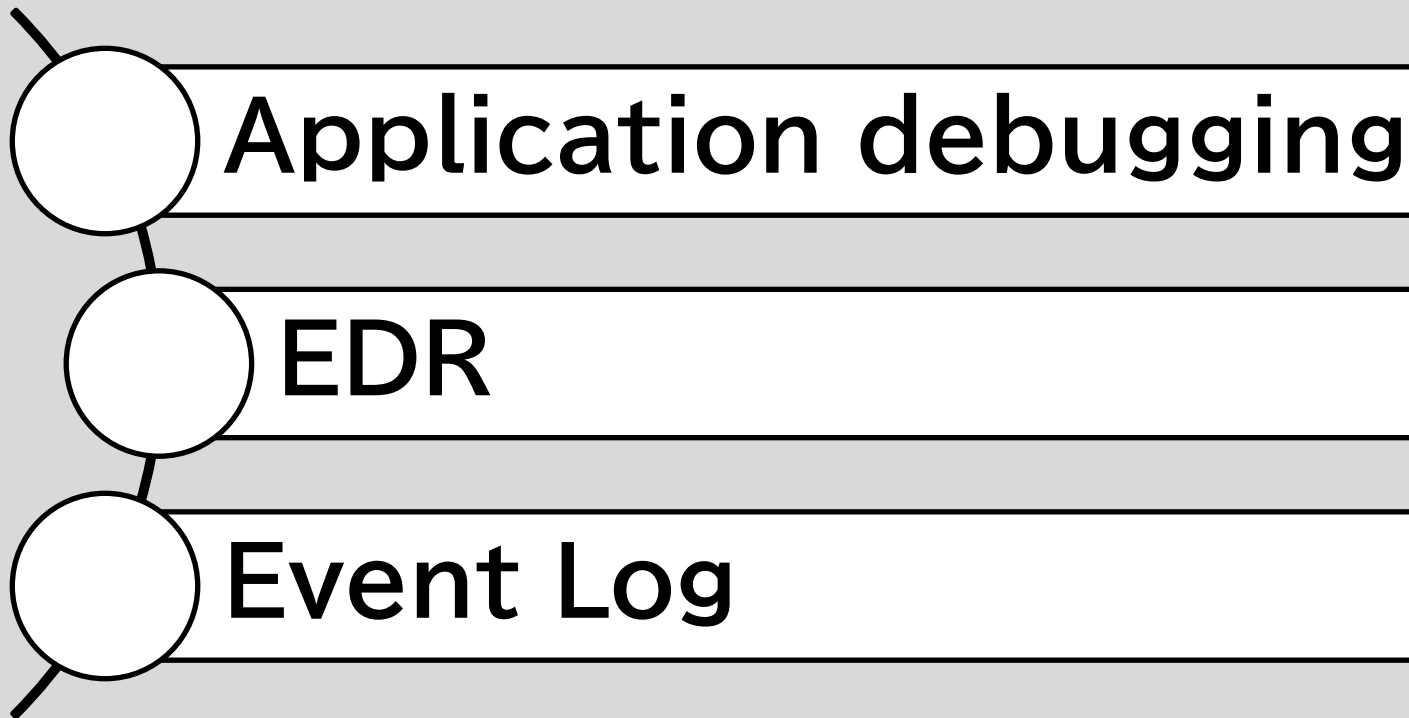
How it Works

 **YAMA Goya** uses ETW event to collect various activities (file, registry, network, powershell, WMI etc.) within Windows OS and detect attack activities.

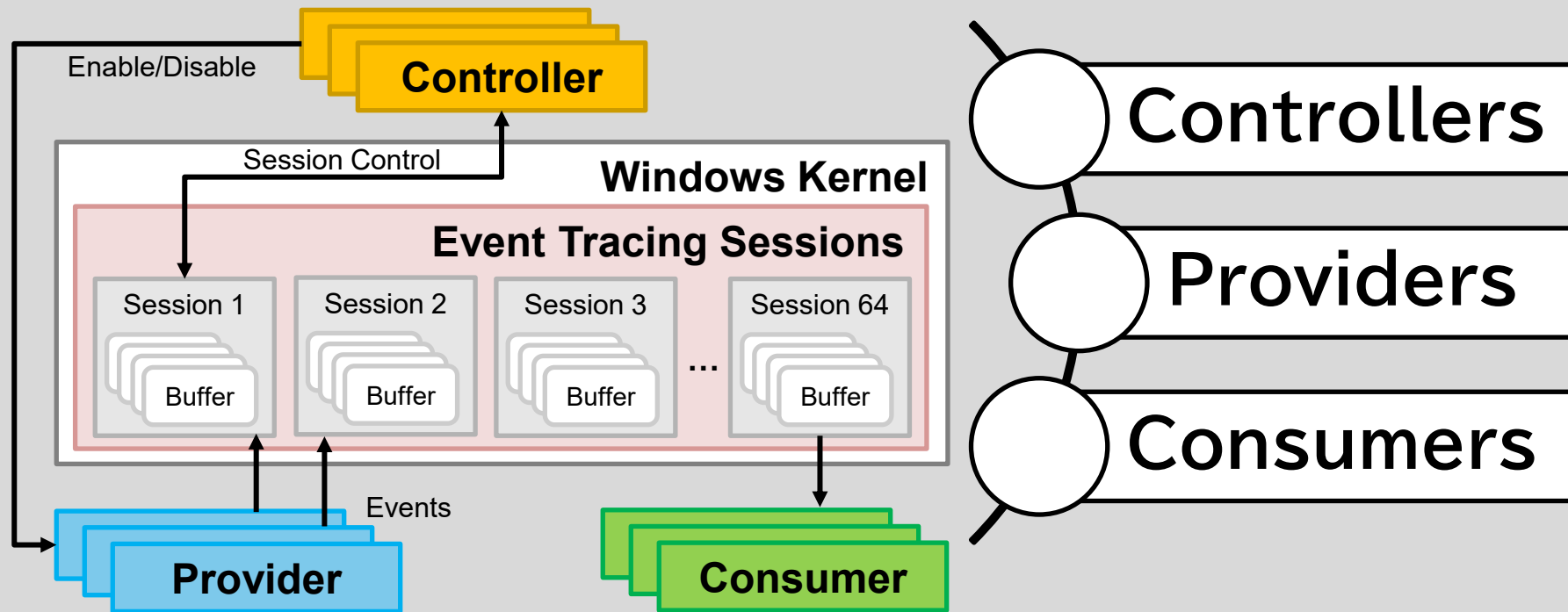
What is Event Tracing for Windows

Event Tracing for Windows is an efficient kernel-level tracing facility that lets you log kernel or application-defined events to a log file. You can consume the events in real time or from a log file and use them to debug an application or to determine where performance issues are occurring in the application.

Use Case for ETW

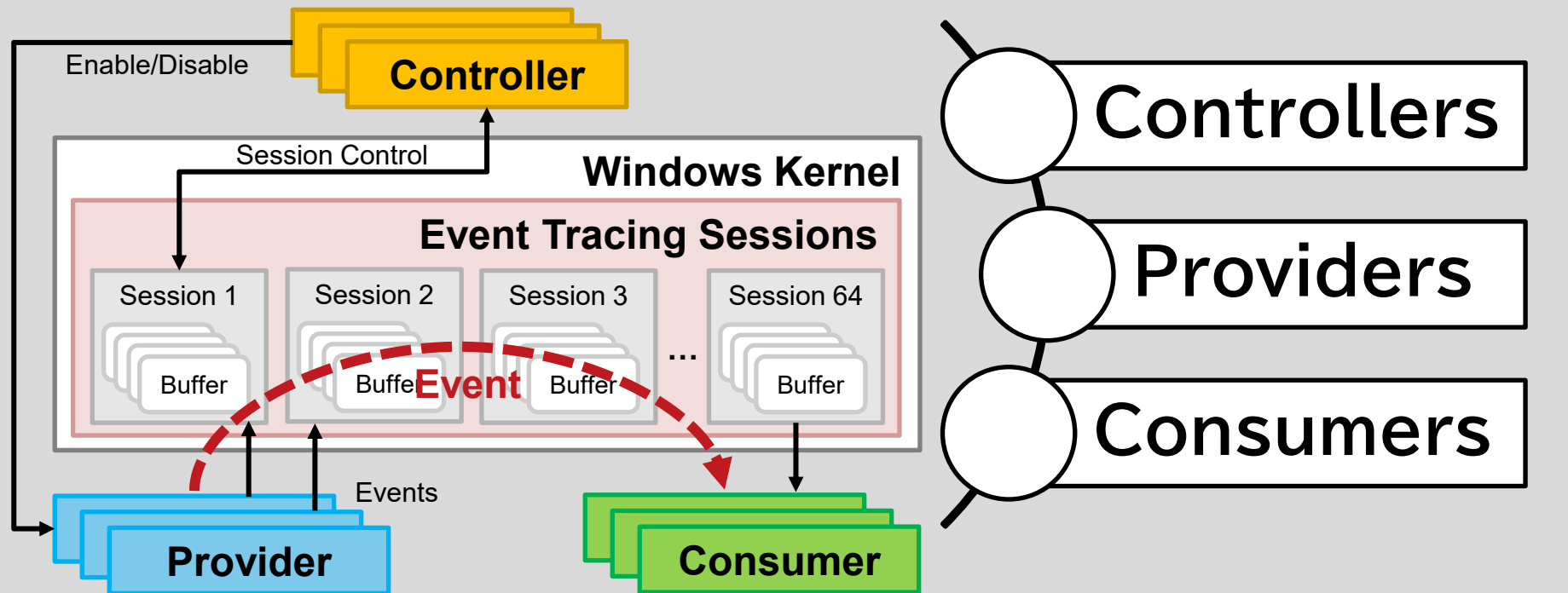


What is Event Tracing for Windows(ETW)



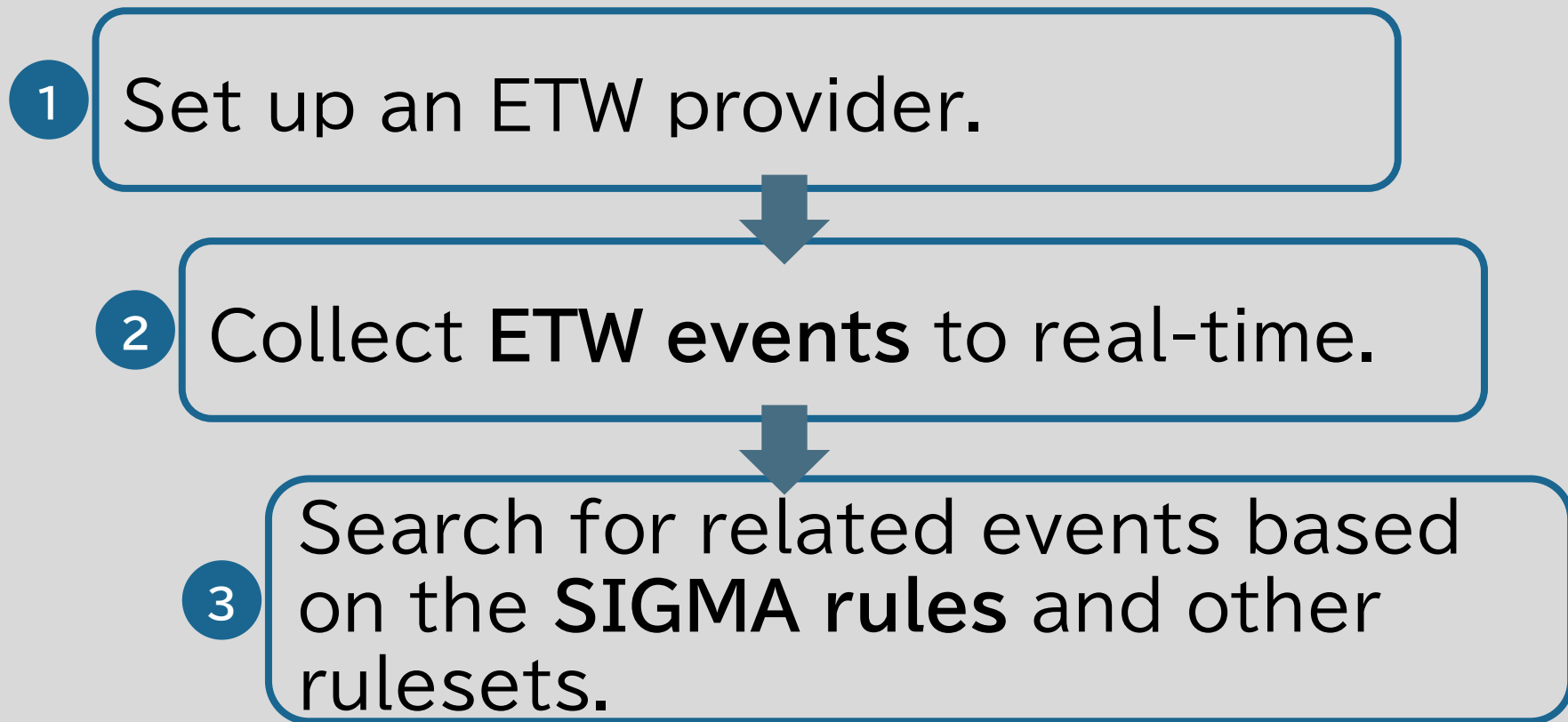
<https://learn.microsoft.com/ja-jp/windows/win32/etw/about-event-tracing>

What is Event Tracing for Windows(ETW)



<https://learn.microsoft.com/ja-jp/windows/win32/etw/about-event-tracing>

How it Works



ETW Providers used by AV/EDR

Research AV/EDR

- ❑ Windows Defender
- ❑ Kaspersky
- ❑ ESET
- ❑ TrendMicro
- ❑ Symantec
- ❑ McAfee
- ❑ Cylance
- ❑ Filebeat (Elastic)

ETW Providers used by AV/EDR

Research AV/EDR

- ❑ Windows Defender
- ❑ Kaspersky
- ❑ ESET
- ❑ TrendMicro
- ❑ Symantec
- ❑ McAfee
- ❑ Cylance
- ❑ Filebeat (Elastic)



Microsoft-Windows-AppModel-Runtime
Microsoft-Windows-AppxPackagingOM
Microsoft-Windows-CAPI2
Microsoft-Windows-Crypto-NCrypt
Microsoft-Windows-Deplorch
Microsoft-Windows-DNS-Client
Microsoft-Windows-Eventlog
Microsoft-Windows-KnownFolders
Microsoft-Windows-LDAP-Client
Microsoft-Windows-NetworkProfile
Microsoft-Windows-Perflib
Microsoft-Windows-PrintService
Microsoft-Windows-RestartManager
Microsoft-Windows-RPC-Events
Microsoft-Windows-Shell-Core
Microsoft-Windows-User Profiles General
Microsoft-Windows-UserPnp
Microsoft-Windows-VerifyHardwareSecurity
Microsoft-Windows-WinHttp
Microsoft-Windows-WinINet-Pca
Network Location Awareness Trace
Network Profile Manager
WLAN Diagnostics Trace

ETW Providers used by AV/EDR

Research AV/EDR

- ❑ Windows Defender
- ❑ Kaspersky
- ❑ ESET
- ❑ TrendMicro
- ❑ Symantec
- ❑ McAfee
- ❑ Cylance
- ❑ Filebeat (Elastic)



Microsoft-Windows-AppModel-Runtime
Microsoft-Windows-CAPI2
Microsoft-Windows-Crypto-NCrypt
Microsoft-Windows-Deplorch
Microsoft-Windows-DNS-Client
Microsoft-Windows-Eventlog
Microsoft-Windows-Kernel-AppCompat
Microsoft-Windows-KnownFolders
Microsoft-Windows-LDAP-Client
Microsoft-Windows-Networking-Correlation
Microsoft-Windows-RPC
Microsoft-Windows-RPC-Events
Microsoft-Windows-Shell-Core
Microsoft-Windows-User Profiles General
Microsoft-Windows-UserPnp
Microsoft-Windows-WinHttp
Microsoft-Windows-WinINet-Pca

ETW Providers used by AV/EDR

Important ETW Provider List

Microsoft-Windows-AppModel-Runtime	Microsoft-Windows-PrintService
Microsoft-Windows-CAPI2	Microsoft-Windows-RPC
Microsoft-Windows-COMRuntime	Microsoft-Windows-RPC-Events
Microsoft-Windows-Crypto-NCrypt	Microsoft-Windows-Shell-Core
Microsoft-Windows-Deplorch	Microsoft-Windows-User Profiles General
Microsoft-Windows-DNS-Client	Microsoft-Windows-UserPnp
Microsoft-Windows-Eventlog	Microsoft-Windows-VerifyHardwareSecurity
Microsoft-Windows-Kernel-AppCompat	Microsoft-Windows-WinHttp
Microsoft-Windows-KnownFolders	Microsoft-Windows-WinINet-Pca
Microsoft-Windows-LDAP-Client	Network Profile Manager
Microsoft-Windows-Networking-Correlation	WLAN Diagnostics Trace
Microsoft-Windows-NetworkProfile	

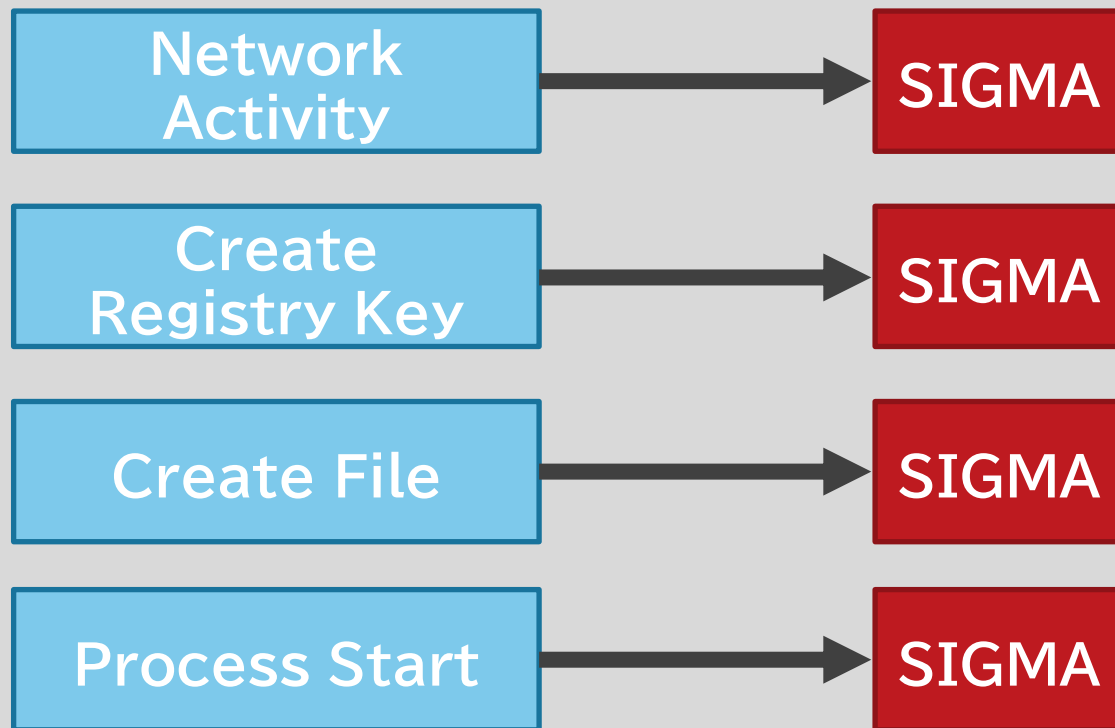
ETW Providers used by YAMAGoya

ETW Provider	ID	Details
Microsoft-Windows-Kernel-File	12,15	Create, Read
Microsoft-Windows-Kernel-Process	1	ProcessStart
Microsoft-Windows-Kernel-Registry	1,2,5,7	CreateKey, OpenKey, SetValueKey, QueryValueKey
Microsoft-Windows-DNS-Client	3000-3020	Query
Microsoft-Windows-Kernel-Network	-	TCP/IP
Microsoft-Windows-PowerShell	4104	StartingCommand
Microsoft-Windows-Shell-Core	9707, 28115	ExecutingFromRunKeyAsJobStart, AddShortcut
Microsoft-Windows-WinINet-Capture	-	WinINet
Microsoft-Windows-WinRM	6, 91	WSManSessioninitializeStart, WinRM connection
Microsoft-Windows-WMI-Activity	11	WMI connection

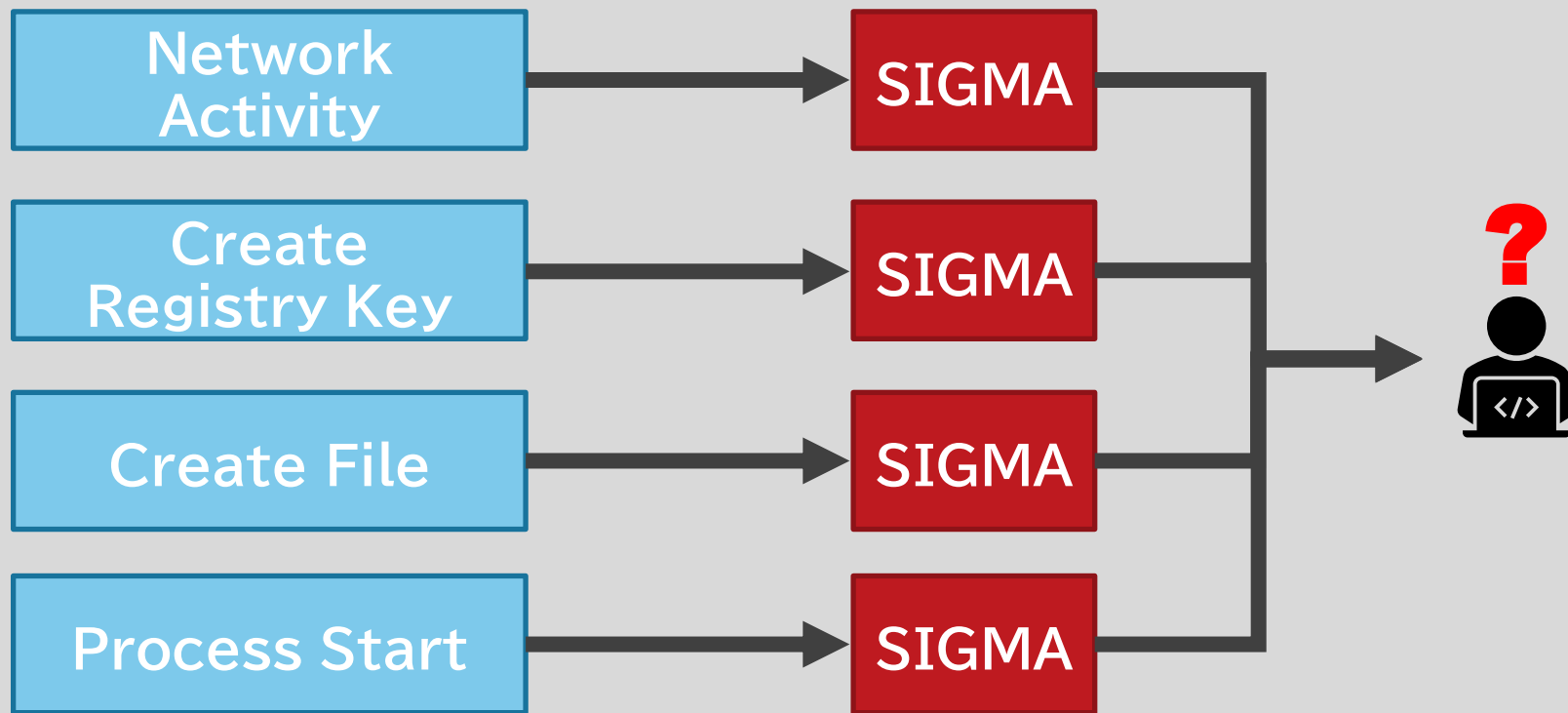
ETW Providers used by YAMAGoya

ETW Provider	ID	Details
Microsoft-Windows-Security-Mitigations	-	https://learn.microsoft.com/ja-jp/defender-endpoint/exploit-protection
Microsoft-Windows-Security-Adminless	-	Adminless mode
Microsoft-Windows-Security-LessPrivilegedAppContainer	-	
Microsoft-Windows-Audit-CVE	-	
Microsoft-Windows-SMBServer	552, 650, 700	Smb2SessionAuthenticated, Smb2FileOpenStart, Smb2ShareAdd
Microsoft-Windows-SMBClient	31010, 30501, 30508	Authorization,
Microsoft-Windows-Kernel-EventTracing	2, 3, 10, 11	ETW session start / stop

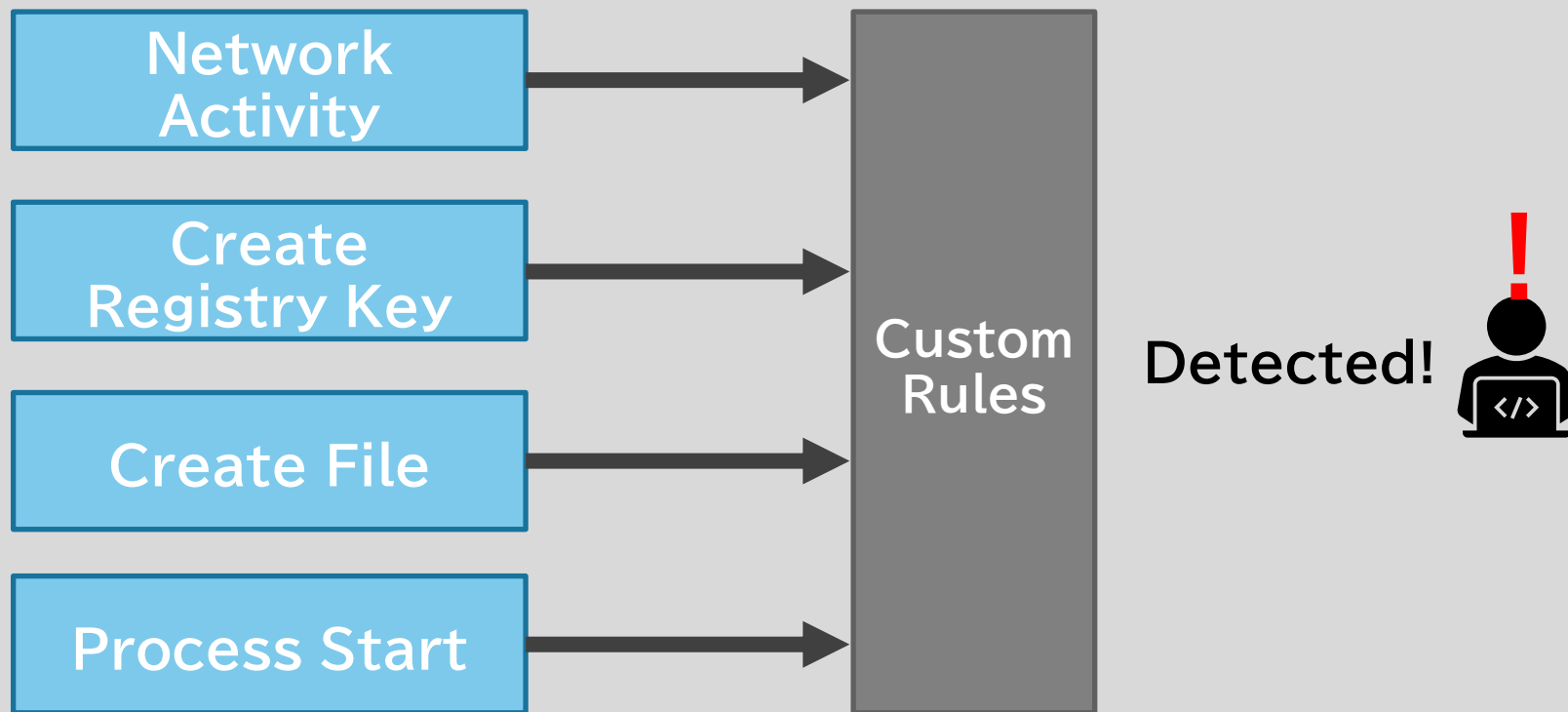
Advanced Attack Behavior Analysis



Advanced Attack Behavior Analysis

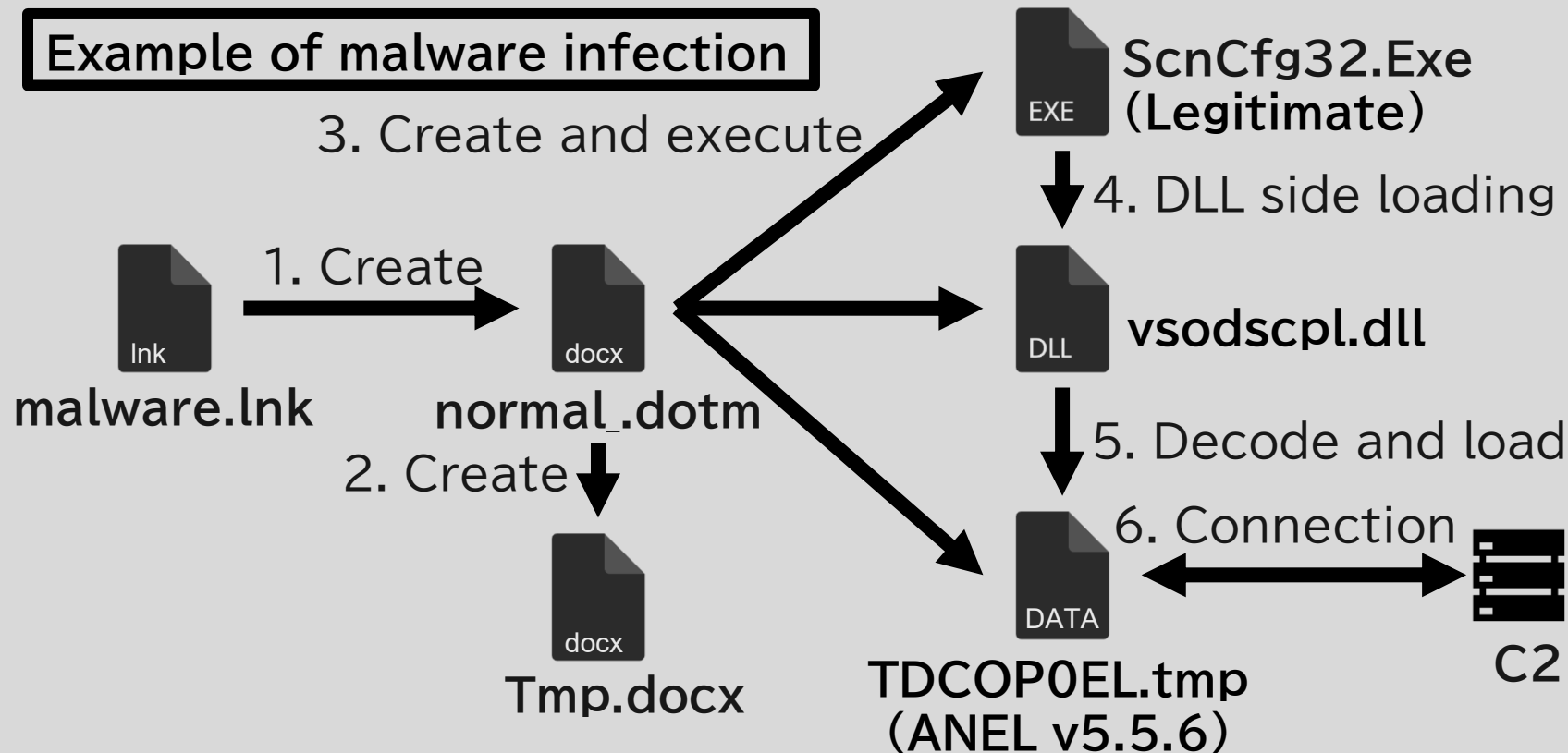


Advanced Attack Behavior Analysis



How to Write YAMAGoya Custom Rule

Example of malware infection



How to Write YAMAGoya Custom Rule

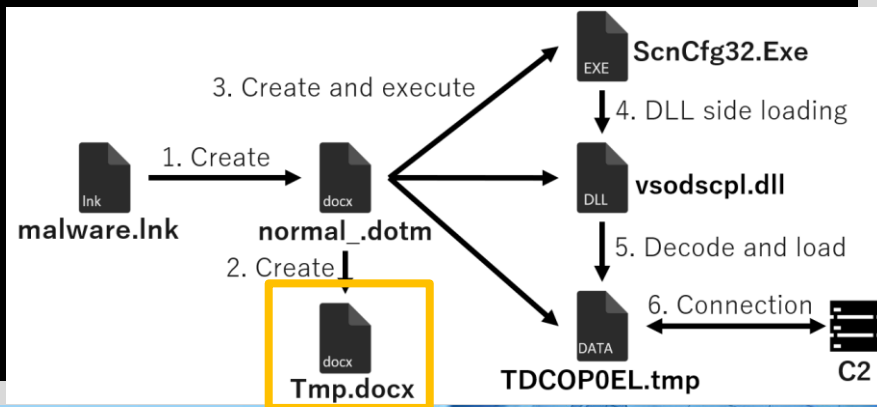
```
rulename: "ANEL"
description: "Detects ANEL from maldoc type"
rules:
  - ruletype: "regex"
    target: "file"
    rule: "Tmp¥¥.docx$"
  - ruletype: "regex"
    target: "process"
    rule: "ScnCf32¥¥.Exe$"
  - ruletype: "regex"
    target: "dll"
    rule: "vsodscpl¥¥.dll$"
  - ruletype: "regex"
    target: "file"
    rule: "TCDolW0p¥¥.log$"
  - ruletype: "ipv4"
    target: "ipv4"
    rule: "45.32.116.146"
```

How to Write YAMAGoya Custom Rule

```
rulename: "ANEL"  
description: "Detects ANEL from maldoc type"  
rules:
```

- ruletype: "regex"
target: "file"
rule: "Tmp¥¥.docx\$"
- ruletype: "regex"
target: "process"
rule: "ScnCfg32¥¥.Exe\$"
- ruletype: "regex"
target: "dll"
rule: "vsodscpl¥¥.dll\$"
- ruletype: "regex"
target: "file"
rule: "TCDolW0p¥¥.log\$"
- ruletype: "ipv4"
target: "ipv4"
rule: "45.32.116.146"

→ Decoy file

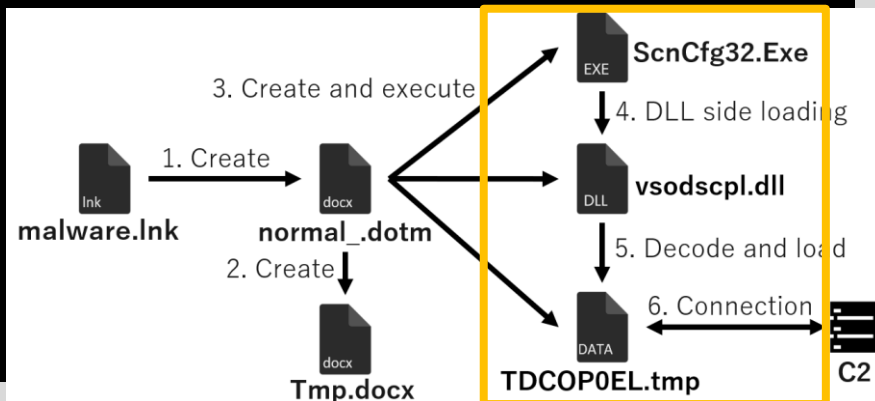


How to Write YAMAGoya Custom Rule

```
rulename: "ANEL"  
description: "Detects ANEL from maldoc type"  
rules:
```

- ruletype: "regex"
target: "file"
rule: "Tmp¥¥.docx\$"
- ruletype: "regex"
target: "process"
rule: "ScnCf32¥¥.Exe\$"
- ruletype: "regex"
target: "dll"
rule: "vsodscpl¥¥.dll\$"
- ruletype: "regex"
target: "file"
rule: "TCDolW0p¥¥.log\$"
- ruletype: "ipv4"
target: "ipv4"
rule: "45.32.116.146"

DLL side-loading of malware.

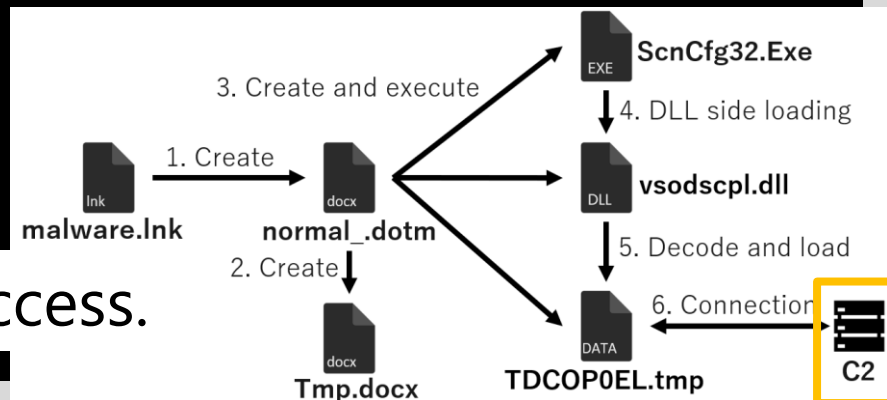


How to Write YAMAGoya Custom Rule

```
rulename: "ANEL"  
description: "Detects ANEL from maldoc type"  
rules:
```

- ruletype: "regex"
target: "file"
rule: "Tmp¥¥.docx\$"
- ruletype: "regex"
target: "process"
rule: "ScnCf32¥¥.Exe\$"
- ruletype: "regex"
target: "dll"
rule: "vsodscpl¥¥.dll\$"
- ruletype: "regex"
target: "file"
rule: "TCDolW0p¥¥.log\$"
- ruletype: "ipv4"
target: "ipv4"
rule: "45.32.116.146"

C2 access.



How to Write YAMAGoya Custom Rule

```
rulename: "ANEL"  
description: "Detects ANEL from maldoc type"  
rules:
```

- ruletype: "regex"
target: "file"
rule: "Tmp¥¥.docx\$"
- ruletype: "regex"
target: "process"
rule: "ScnCf32¥¥.Exe\$"
- ruletype: "regex"
target: "dll"
rule: "vsodscpl¥¥.dll\$"
- ruletype: "regex"
target: "file"
rule: "TCDolW0p¥¥.log\$"
- ruletype: "ipv4"
target: "ipv4"
rule: "45.32.116.146"

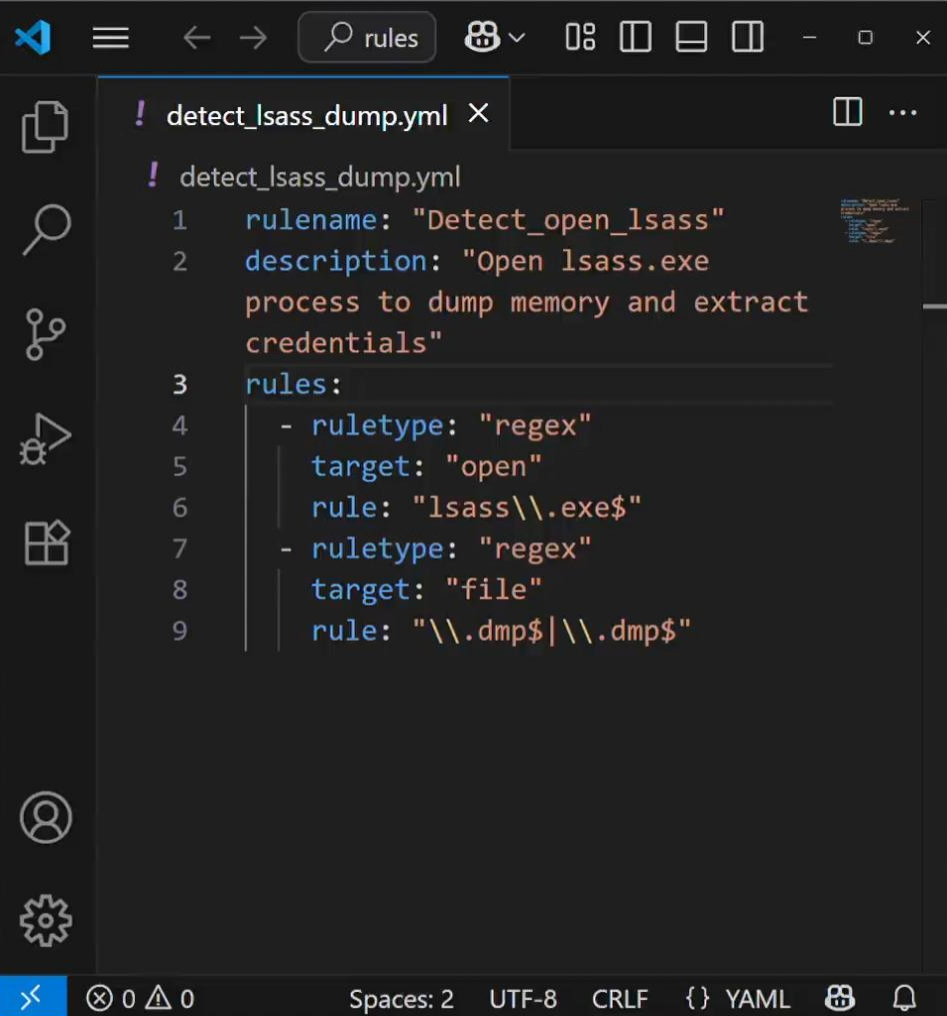


YAMAGoya detects this malware infection activity if it is checked within the time slot (default 10 seconds).

Demo

- Behavioral detection using
Custom YAML rule -

C:\YAMAGoya>



```
! detect_lsass_dump.yml X
! detect_lsass_dump.yml
1  rulename: "Detect_open_lsass"
2  description: "Open lsass.exe
   process to dump memory and extract
   credentials"
3  rules:
4    - ruletype: "regex"
5      target: "open"
6      rule: "lsass\\.exe$"
7    - ruletype: "regex"
8      target: "file"
9      rule: "\\\\.dmp$|\\.dmp$"
```

Use Case for Custom Rules (Demo)

Stealing credentials from lsass

```
rulename: "Detect_open_lsass"
description: "Open lsass.exe process to dump
memory and extract credentials"
rules:
  - ruletype: "regex"
    target: "open"
    rule: "lsass¥¥.exe$"
  - ruletype: "regex"
    target: "file"
    rule: "¥¥.dmp$|¥¥.dmp$"
```

Target Activity for Custom Rules

Target Activity			
File	File creation events.	DNS	DNS events.
Delfile	File deletion events.	IPv4	IPv4 network events.
Process	Process events.	IPv6	IPv6 network events.
Open	OpenProcess.	Shell	Shell-related events.
Load	DLL load events.	PowerShell	PowerShell execution events.
Registry	Registry events.	WMI	WMI command execution events.

Use Case for Custom Rules 1

Ransomware Decoy File Detection Technique

```
rulename: "Decoy_file_deleted"
description: "Detecting ransomware using decoy files"
rules:
  - ruletype: "regex"
    target: "delfile"
    rule: "decoy¥¥.docx$"
  - ruletype: "regex"
    target: "delfile"
    rule: "decoy¥¥.png$"
```

Use Case for Custom Rules 2

Detecting WMI Activity used by Malware

```
rulename: "WMI_command_activity"
description: "Detects WMI commands using attacker"
rules:
  - ruletype: "regex"
    target: "wmi"
    rule: "SELECT ¥¥* FROM Win32_NetworkAdapterConfiguration"
  - ruletype: "regex"
    target: "wmi"
    rule: "SELECT ¥¥* FROM Win32_OperatingSystem"
  - ruletype: "regex"
    target: "wmi"
    rule: "SELECT ¥¥* FROM Win32_ComputerSystem"
```


Use Case for Custom Rules 3

Detection of PowerShell Activity

```
rulename: "PowerShell_command_activity"
description: "Detects Invoke-SharpLoader.ps1"
rules:
  - ruletype: "regex"
    target: "powershell"
    rule: "MemoryStream"
  - ruletype: "regex"
    target: "powershell"
    rule: "GZipStream"
  - ruletype: "regex"
    target: "powershell"
    rule: "EntryPoint.Invoke"
```



shu-tom Upload ✓

Preview

Code

Blame

499 lines (386 loc) · 21.2 KB

Raw Download Edit



Concept

YAMAGoya (Yet Another Memory Analyzer for malware detection and Guarding Operations with YARA and SIGMA) is a C# application that leverages [Event Tracing for Windows \(ETW\)](#) to capture real-time system events. It applies detection rules written in YAML format (for custom correlation logic) and can also parse **SIGMA** rules for standardized threat detection. In addition, it supports in-memory scanning using **YARA** to detect fileless or stealth malware.

The tool run: <https://github.com/JPCERTCC/YAMAGoya/> itures.

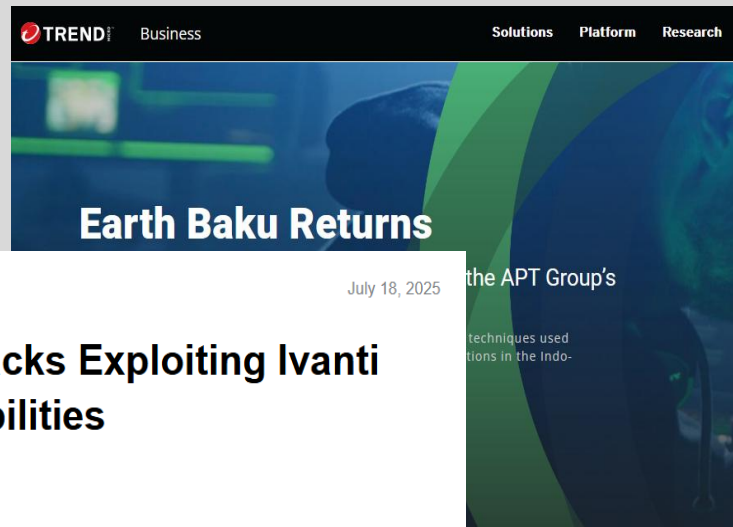
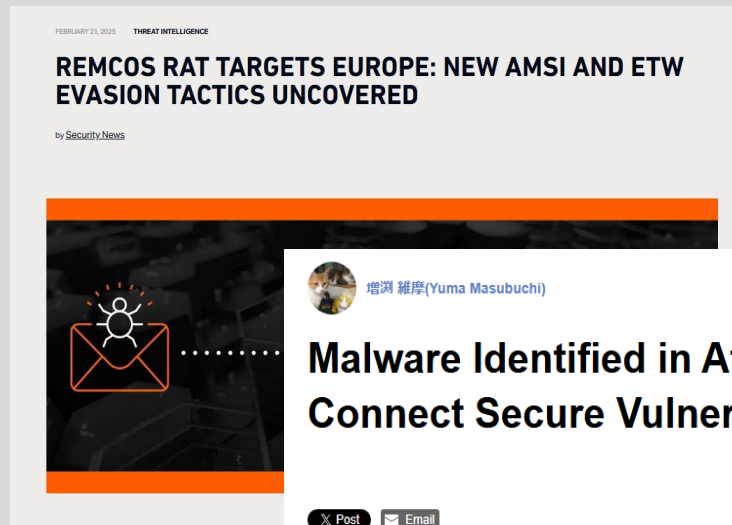
Table of Contents

Weakness of YAMAGoya



YAMAGoya might not detect properly malicious behavior when an attacker uses **ETW bypass** techniques

Campaigns using ETW bypass



増淵 雅彦(Yuma Masubuchi) July 18, 2025

Malware Identified in Attacks Exploiting Ivanti Connect Secure Vulnerabilities

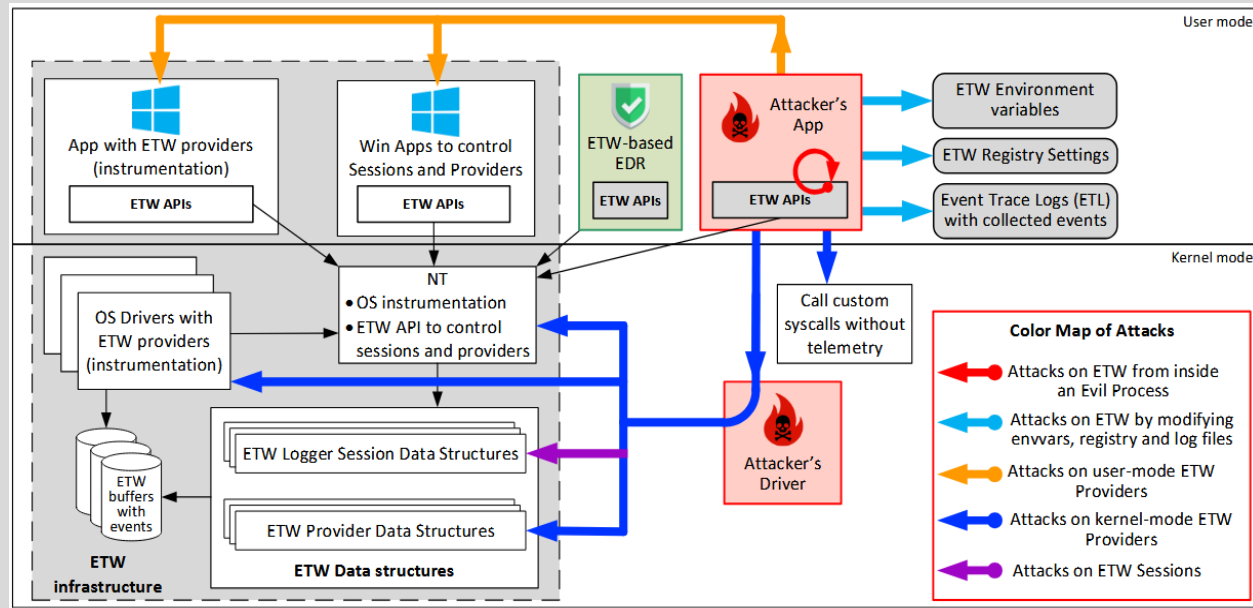
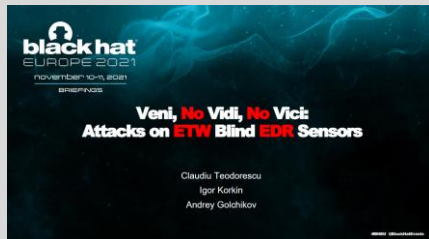
Post Email

JPCERT/CC Eyes previously introduced the malware **SPAWNCHIMERA** and **DslogdRAT**, which were deployed by exploiting vulnerabilities in Ivanti Connect Secure. At JPCERT/CC, we have continued to observe active exploitation of these vulnerabilities. In this report, we explain the following malware, tools, and penetration tactics used by attackers leveraging **CVE-2025-0282** and **CVE-2025-22457** in attacks observed from December 2024 to the present, July 2025.

- MDifyLoader and Cobalt Strike Beacon
- vshell
- Fscan

https://blogs.jpcert.or.jp/en/2025/07/ivanti_cs.html
<https://www.sonicwall.com/blog/remcos-rat-targets-europe-new-amsi-and-etw-evasion-tactics-uncovered>
<https://www.trendmicro.com/vinfo/us/security/news/cybercrime-and-digital-threats/earth-baku-returns>

Attack surfaces of ETW



<https://i.blackhat.com/EU-21/Wednesday/EU-21-Teodorescu-Veni-No-Vidi-No-Vici-Attacks-On-ETW-Blind-EDRs.pdf>

Detectable ETW bypass by YAMAGoya



Config modification
via PowerShell



Registry
modification



ETL file deletion

```
rulename "KillETW"
description "Detects KillETW.ps1"
rules
  ruletype "regex"
  target: "powershell"
  rule: "System.Diagnostics.Eventing.EventProvider"
  - ruletype: "regex"
  target: "powershell"
  rule:
    "System.Management.Automation.Tracing.PSEtwLogProvider"
  - ruletype: "regex"
  target: "powershell"
  rule: "SetValue"
```

```
C:\YAMAGoya>YAMAGoya.exe -s -d rules -a
```

```
C:\YAMAGoya>Running in command-line mode...
```

```
[INFO] Using default session name: YAMAGoya
```

```
Starting ETW session...
```

```
ETW session started. Press Ctrl+C to stop.
```

```
[INFO] Loading YAML rule files...
```

```
[INFO] Loading detection rules from C:\YAMAGoya\rules\detect_invoke-sharploader.yml
```

```
[INFO] Loading detection rules from C:\YAMAGoya\rules\detect_killetw.yml
```

```
[INFO] Loading detection rules from C:\YAMAGoya\rules\detect_isass_dump.yml
```

```
[INFO] Loading detection rules from C:\YAMAGoya\rules\detect_malware_sample.yml
```

```
[INFO] Loading detection rules from C:\YAMAGoya\rules\detect_ransomware_using_decoy.yml
```

```
[INFO] Loading detection rules from C:\YAMAGoya\rules\detect_wmi_activity.yml
```

```
[INFO] Successfully loaded 6 rules. (Skipped 0 files with errors)
```

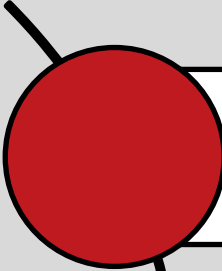
```
[INFO] Loaded default rule count: 6
```

```
[INFO] Press Ctrl+C to stop.
```

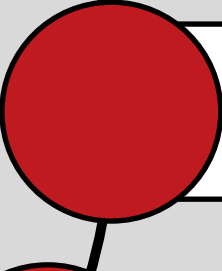
```
[INFO] Starting ETW detection (check interval: 10s)...
```

```
[INFO] 2025/07/25 11:51:39 DETECTED: KillETW - Detects KillETW.ps1
```

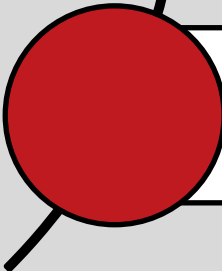
Undetectable ETW bypass by YAMAGoya



Patching user mode
ETW provider



Attack on kernel
mode ETW provider



Attack on ETW
session

```
void DisableETW(void) {
    DWORD oldprotect = 0;

    unsigned char sEtwEventWrite[] = { 'E','t','w','E','v','e','n','t','W','r','i','t','e', 0x0 };

    // change the protection so we can write to it
    void* pEventWrite = GetProcAddress(GetModuleHandle(L"ntdll.dll"), (LPCSTR)sEtwEventWrite);

    VirtualProtect_p(pEventWrite, 4096, PAGE_EXECUTE_READWRITE, &oldprotect);

#ifdef _WIN64
    // the return from this function will be 0
    memcpy(pEventWrite, "\x48\x33\xc0\xc3", 4); // xor rax, rax; ret
#else
    memcpy(pEventWrite, "\x33\xc0\xc2\x14\x00", 5); // xor eax, eax; ret 14
#endif

    VirtualProtect_p(pEventWrite, 4096, oldprotect, &oldprotect);
    FlushInstructionCache(GetCurrentProcess(), pEventWrite, 4096);
}
```

Demo

- ETW bypass detection -

C:\YAMAGoya>

```
Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Message Learn More
Detect_bypass_etw.yml
C:\YAMAGoya> rules > ! Detect_bypass_etw.yml
1 rulename: "BypassETW"
2 description: "Detects 8KuV.ps1 in the campaign of REMCOS RAT t
3 rules:
4   - ruletype: "regex"
5     target: "powershell"
6     rule: "0x45,0x74,0x77,0x45,0x76,0x65,0x6E,0x74,0x57,0x72,0
7   - ruletype: "regex"
8     target: "powershell"
9     rule: "0xC3"
10  - ruletype: "regex"
11    target: "powershell"
12    rule: "0xb8,0xff,0x55"
13  - ruletype: "regex"
14    target: "powershell"
15    rule: "[System.Text.Encoding]::ASCII.GetString"
16
```

Custom Rules (Demo2)

Ransomware Decoy File Detection Technique

rulename "BypassETW"
description "Detects 8KuV.ps1 in the campaign of REMCOS RAT targets Europe"
rules

- ruletype: "regex"
target: "powershell"

rule: "0x45,0x74,0x77,0x45,0x76,0x65,0x6E,0x74,0x57,0x72,0x69,0x74,0x65"

ruletype "regex"
target: "powershell"

rule: "0xC3"

- ruletype "regex"
target: "powershell"

rule: "0xb8,0xff,0x55"

- ruletype "regex"
target: "powershell"

rule: "[System.Text.Encoding]::ASCII.GetString"

"EtwEventWrite"

```
$SetwFuncName = "EtwEventWrite"
$SetwAddr = Get-SystemProcAddress ("ntdll.dll") $SetwFuncName
if ($SetwAddr -eq $null) {
    Throw "[!] Erro ao obter o endereço de $SetwFuncName" #Error getting address of $SetwFuncName
}

if (-not $memProtectDelegate.Invoke($SetwAddr, 1, $PAGE_EXECUTE_WRITECOPY, [ref]$origProt)) {
    Throw "[!] Erro ao alterar a proteção de memória em $SetwFuncName" #Error changing memory protection on $SetwFuncName
}

try {
    if ($PtrSize -eq 8) {
        $marshal::WriteByte($SetwAddr, 0xC3)
    }
    else {
        $SetwPatch = [byte[]] (0xb8, 0xff, 0x55)
        $marshal::Copy($SetwPatch, 0, [IntPtr]$SetwAddr, 3)
    }
}
```

Patching EtmEventWrite by
Either Ret[0xC3]
Or Mov eax,00000NAN[0xb8,0xff,0x55]

<https://www.sonicwall.com/blog/remcos-rat-targets-europe-new-amsi-and-etw-evasion-tactics-uncovered>

FAQ

Q1: Can YAMAGoya replace traditional antivirus software?

A: **No.** YAMAGoya is designed to complement antivirus software by focusing on behavioral and memory-based detection, especially for fileless malware and advanced attacks.

Q2: Can YAMAGoya run continuously in the background?

A: **Yes.** YAMAGoya supports system tray integration and can run quietly in the background while actively monitoring for threats using the configured rules.

Q3: Can YAMAGoya be integrated into an existing SIEM platform?

A: **Yes.** Although YAMAGoya is a standalone tool, it outputs logs in text and Windows Event Log formats. These can be forwarded to SIEMs like Splunk using standard log collectors or agents.

Q4: Are there any known limitations regarding ETW Bypass?

A: **Yes.** YAMAGoya currently does not implement specific countermeasures against ETW bypass techniques. This means advanced attackers may attempt to disable or tamper with ETW to avoid detection. We recommend using YAMAGoya in combination with other monitoring solutions for a more robust defense.



Thank you!

 @jpcert_en
@jpcert_ac

 ir-info@jpcert.or.jp
PGP <https://www.jpcert.or.jp/english/pgp/>